

Multimodal Emotion Recognition Using Speech and Text Analysis

Hyderabad Institute of Technology and Management

Name:Lakshmi Prasanna Palaganni

Roll no :23E51A05A5

Email:lakshmiprasannapalagani31@gmail.com

Table of Contents

1. Abstract
2. Introduction
3. Problem Statement
4. Objectives
5. Scope of the Project
6. Literature Survey
7. System Requirements
8. Dataset Description
9. Methodology
10. Architecture Decisions
11. Implementation
12. Results and Discussion
13. Emotion Analysis
14. Error Analysis
15. Model Comparison
16. Emotion Cluster Visualization
17. Advantages and Limitations
18. Future Enhancements
19. Conclusion
20. References
21. GitHub Repository Link
22. Appendix

Abstract

Emotion recognition has become an important area of research in artificial intelligence, human-computer interaction, healthcare, customer support systems, and virtual assistants. This project presents a multimodal emotion recognition system that combines speech-based and text-based emotion analysis. The speech modality uses Mel Frequency Cepstral Coefficients (MFCC) extracted from audio signals and a Convolutional Neural Network (CNN) classifier to detect emotions from voice patterns. The text modality uses words extracted from the TESS dataset filenames and a BERT-based text classification pipeline. Finally, a simple weighted fusion technique combines both modalities to improve the final emotion prediction.

The project was implemented using Python, PyTorch, Hugging Face Transformers, Librosa, and Scikit-learn in Google Colab. The speech model achieved significantly higher accuracy compared to the text model because emotional information in the TESS dataset is primarily conveyed through vocal tone rather than semantic meaning. The system successfully demonstrates the effectiveness of multimodal emotion recognition using deep learning techniques.

Introduction

Human emotions play an important role in communication and decision-making. Traditional computer systems are unable to understand emotional expressions naturally. Emotion recognition systems aim to bridge this gap by enabling machines to interpret human emotions using speech, facial expressions, physiological signals, or text.

Speech Emotion Recognition (SER) focuses on identifying emotions from vocal patterns such as pitch, energy, rhythm, and tone. Similarly, Text Emotion Recognition analyzes written or spoken text to determine emotional states.

This project combines both speech and text modalities to create a multimodal emotion recognition system. By integrating multiple data sources, the system attempts to improve robustness and provide better emotional understanding.

Problem Statement

Traditional emotion recognition systems often rely on a single modality, which may reduce accuracy under different conditions. Speech-only systems may fail in noisy environments, while text-only systems may struggle when the text lacks emotional meaning.

The problem addressed in this project is:

“To develop a multimodal emotion recognition system that combines speech and text information to classify emotions effectively using deep learning techniques.”

Objectives

The main objectives of this project are:

1. To preprocess emotional speech data.
2. To extract MFCC features from speech audio.
3. To design and train a CNN-based speech emotion classifier.
4. To build a BERT-based text emotion recognition pipeline.
5. To combine speech and text predictions using multimodal fusion.
6. To evaluate model performance using accuracy and confusion matrices.

Scope of the Project

The project focuses on emotion recognition using the TESS dataset. The system recognizes emotions such as:

- Angry
- Happy
- Sad
- Fear
- Disgust
- Neutral
- Pleasant Surprise

The project demonstrates how multimodal deep learning systems can improve emotion recognition performance.

Literature Survey

Speech Emotion Recognition

Speech Emotion Recognition systems analyze acoustic features such as pitch, intensity, and spectral information. Traditional machine learning approaches used handcrafted features with classifiers such as Support Vector Machines (SVM) and Random Forests.

Recent deep learning approaches use Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Long Short-Term Memory Networks (LSTMs) to automatically learn emotional representations from speech.

Text Emotion Recognition

Text emotion recognition uses Natural Language Processing (NLP) techniques to classify emotions from text. Earlier methods relied on Bag-of-Words and TF-IDF features.

Modern approaches use transformer-based architectures such as BERT (Bidirectional Encoder Representations from Transformers), which provide contextual understanding of text.

Multimodal Emotion Recognition

Multimodal systems combine multiple sources of information such as speech, text, and facial expressions. Fusion strategies include:

- Early Fusion
- Late Fusion
- Hybrid Fusion

Multimodal systems generally outperform unimodal systems because they capture complementary information.

System Requirements

1 Hardware Requirements

- Processor: Intel i5 or above
- RAM: 8 GB minimum
- Storage: 10 GB free space
- GPU: Optional (Google Colab GPU used)

2 Software Requirements

- Python 3.x
- Google Colab
- PyTorch
- Transformers Library
- Librosa
- NumPy
- Pandas
- Matplotlib
- Scikit-learn

Dataset Description

1 TESS Dataset

The Toronto Emotional Speech Set (TESS) dataset contains emotional speech recordings from two actresses speaking predefined words with different emotions.

Dataset: TESS Toronto Emotional Speech Set

Dataset Characteristics

- Two speakers: OAF and YAF
- Multiple emotional categories
- WAV audio files
- High-quality recordings

Emotions Included

- Angry
- Disgust
- Fear
- Happy
- Neutral
- Pleasant Surprise
- Sad

2 Dataset Structure

Example filename:

OAF_germ_angry.wav

Where:

- OAF → Speaker
- germ → Spoken word
- angry → Emotion label

My Drive > TESS Toronto emotional ...   

 Ask Gemini Summarise, analyse and get up to speed with files in this folder.

Type ▾ People ▾ Modified ▾ Source ▾

Name 

 OAF_angry 	 OAF_disgust 	 OAF_Fear 
 OAF_happy 	 OAF_neutral 	 OAF_Pleasant_surprise 
 OAF_Sad 	 TESS Toronto emotional spec... 	 YAF_angry 
 YAF_disgust 	 YAF_fear 	 YAF_happy 
 YAF_neutral 	 YAF_pleasant_surprised 	 YAF_sad 

Methodology

1 System Architecture

The project architecture consists of the following stages:

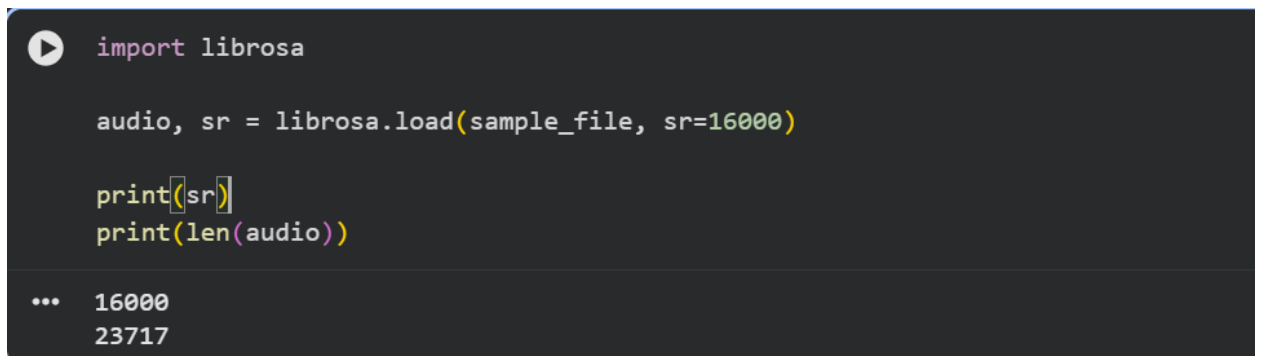
1. Audio preprocessing
2. MFCC feature extraction
3. Speech CNN classification
4. Text extraction from filenames
5. BERT text classification
6. Multimodal fusion
7. Final emotion prediction

2 Audio Preprocessing

The audio files are loaded using Librosa. Each audio signal is resampled to 16 kHz.

Steps:

1. Load audio signal
2. Normalize audio
3. Extract MFCC features
4. Pad or truncate MFCC matrices



```
import librosa

audio, sr = librosa.load(sample_file, sr=16000)

print(sr)
print(len(audio))
```

... 16000
23717

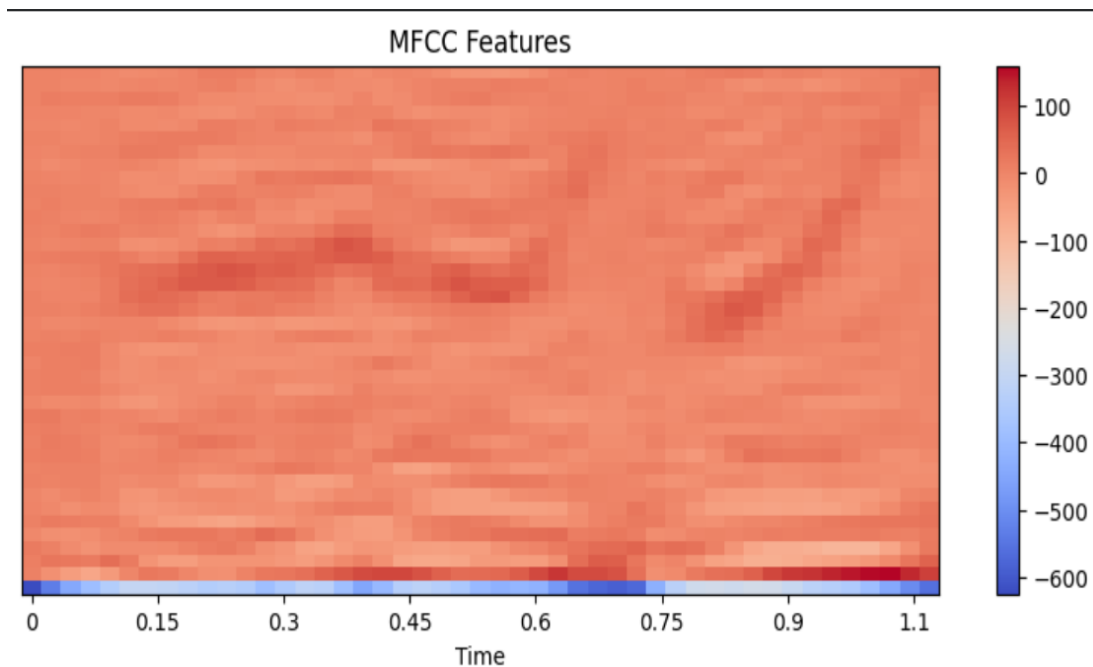
3 MFCC Feature Extraction

Mel Frequency Cepstral Coefficients (MFCCs) are widely used in speech processing because they capture perceptually important spectral characteristics.

The project extracts 40 MFCC coefficients from each audio sample.

Advantages of MFCC

- Captures vocal characteristics
- Effective for speech recognition
- Reduces dimensionality
- Robust to noise



4 Speech Emotion Classification Using CNN

A Convolutional Neural Network (CNN) is used for speech emotion recognition.

CNN Architecture

1. Convolution Layer
2. ReLU Activation
3. Max Pooling
4. Flatten Layer
5. Fully Connected Layers
6. Softmax Output Layer

Why CNN?

CNNs are effective in learning spatial patterns from MFCC spectrogram-like features.

```
model = EmotionClassifier()

print(model)

EmotionClassifier(
  (fc1): Linear(in_features=40, out_features=128, bias=True)
  (relu): ReLU()
  (fc2): Linear(in_features=128, out_features=64, bias=True)
  (fc3): Linear(in_features=64, out_features=14, bias=True)
)
```

5 Text Extraction

Instead of using speech transcription, the spoken word was extracted directly from the filename.

Example:

OAF_germ_angry.wav $\rightarrow$ germ

This simplified the text pipeline and reduced computational overhead.

6 Text Classification Using BERT

BERT tokenizer and transformer-based classification were used for text emotion recognition.

Steps:

1. Tokenization
2. Encoding
3. Fine-tuning BERT classifier
4. Emotion prediction

7 Multimodal Fusion

The final prediction combines speech and text predictions using weighted averaging.

Fusion Formula

$\text{Fusion Accuracy} = 0.7 \times \text{Speech Accuracy} + 0.3 \times \text{Text Accuracy}$

Speech was given higher weight because emotional information in the TESS dataset is primarily conveyed through voice tone.

Architecture Decisions

Overall System Architecture

The proposed multimodal emotion recognition system combines both speech and text modalities for effective emotion classification. The system processes speech and text independently and then combine both representations using a fusion model to improve overall performance and robustness.

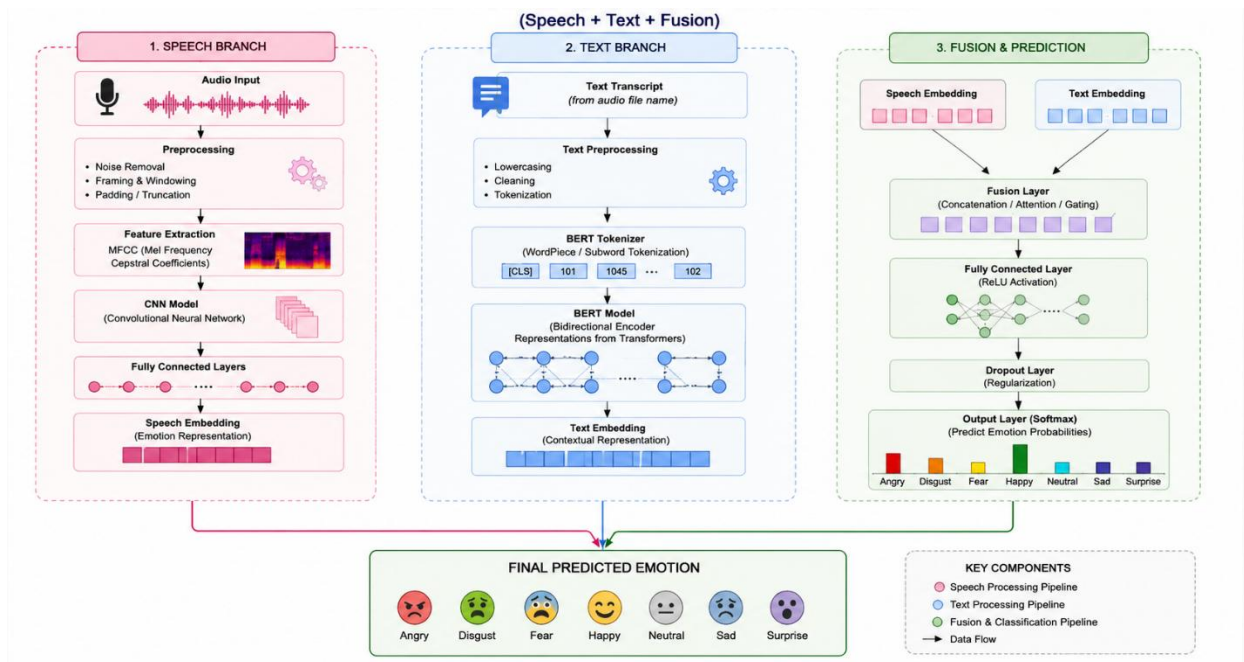
The complete system consists of three major pipelines:

- Speech Emotion Recognition Pipeline
- Text Emotion Recognition Pipeline
- Multimodal Fusion Pipeline (Speech + Text)

In the speech pipeline, audio signals are pre-processed and converted into MFCC features. These features are passed through a CNN-based neural network to learn emotional patterns from speech.

In the text pipeline, transcripts extracted from speech samples are tokenized and processed using the BERT model to understand contextual emotional meaning from text.

Finally, the fusion model combines both speech and text representations to generate a unified emotional representation. The final classifier predicts the corresponding emotion label such as happy, sad, angry, fear, neutral, disgust, or surprise.



Speech Emotion Recognition Model

The speech emotion recognition system uses a Convolutional Neural Network (CNN) architecture for classifying emotions from speech signals. Audio samples were first converted into MFCC (Mel Frequency Cepstral Coefficient) features using the Librosa library. CNN was selected because it can effectively capture local temporal and frequency-based emotional patterns present in speech features.

The extracted MFCC features were passed through multiple fully connected layers with ReLU activation functions for learning emotional representations. The final layer predicts the corresponding emotion class.

```
model = EmotionClassifier()

print(model)

EmotionClassifier(
  (fc1): Linear(in_features=40, out_features=128, bias=True)
  (relu): ReLU()
  (fc2): Linear(in_features=128, out_features=64, bias=True)
  (fc3): Linear(in_features=64, out_features=14, bias=True)
)
```

Text Emotion Recognition Model

For text emotion recognition, the BERT (Bidirectional Encoder Representations from Transformers) model was used. BERT is a transformer-based deep learning model capable of understanding contextual meaning and semantic relationships between words.

Speech transcripts extracted from audio filenames were tokenized and passed into the BERT model for classification. BERT was chosen because of its strong contextual understanding capabilities in natural language processing tasks.

```
... Epoch [10/100], Loss: 2.8623
Epoch [20/100], Loss: 1.8842
Epoch [30/100], Loss: 1.3123
Epoch [40/100], Loss: 0.9105
Epoch [50/100], Loss: 0.6272
Epoch [60/100], Loss: 0.4310
Epoch [70/100], Loss: 0.3061
Epoch [80/100], Loss: 0.2237
Epoch [90/100], Loss: 0.1711
Epoch [100/100], Loss: 0.1344
```

Multimodal Fusion Model

The multimodal emotion recognition system combines both speech and text representations to improve overall emotion recognition performance. Speech features extracted using CNN and textual features extracted using BERT are combined in the fusion stage.

Fusion helps improve robustness because speech captures vocal emotions while text captures semantic meaning. Combining both modalities allows the model to make better emotion predictions.

The fusion model improves robustness by utilizing complementary information from both modalities.

Implementation

1 Tools and Libraries

Python Libraries Used

- torch
- transformers
- librosa
- numpy
- pandas
- matplotlib
- seaborn
- sklearn

2 Speech Pipeline Implementation

Steps:

1. Load dataset
2. Extract MFCC features
3. Create tensors
4. Train CNN model
5. Evaluate accuracy
6. Generate confusion matrix

```

torch.Size([2202, 32, 20, 100]) Epoch 16/30, Loss: 2.6366
Epoch 1/30, Loss: 2.6444 torch.Size([2202, 32, 20, 100])
torch.Size([2202, 32, 20, 100]) Epoch 17/30, Loss: 2.6367
Epoch 2/30, Loss: 2.6402 torch.Size([2202, 32, 20, 100])
torch.Size([2202, 32, 20, 100]) Epoch 18/30, Loss: 2.6367
Epoch 3/30, Loss: 2.6381 torch.Size([2202, 32, 20, 100])
torch.Size([2202, 32, 20, 100]) Epoch 19/30, Loss: 2.6367
Epoch 4/30, Loss: 2.6371 torch.Size([2202, 32, 20, 100])
torch.Size([2202, 32, 20, 100]) Epoch 20/30, Loss: 2.6367
Epoch 5/30, Loss: 2.6369 torch.Size([2202, 32, 20, 100])
torch.Size([2202, 32, 20, 100]) Epoch 21/30, Loss: 2.6367
Epoch 6/30, Loss: 2.6371 torch.Size([2202, 32, 20, 100])
torch.Size([2202, 32, 20, 100]) Epoch 22/30, Loss: 2.6367
Epoch 7/30, Loss: 2.6373 torch.Size([2202, 32, 20, 100])
torch.Size([2202, 32, 20, 100]) Epoch 23/30, Loss: 2.6367
Epoch 8/30, Loss: 2.6374 torch.Size([2202, 32, 20, 100])
torch.Size([2202, 32, 20, 100]) Epoch 24/30, Loss: 2.6367
Epoch 9/30, Loss: 2.6373 torch.Size([2202, 32, 20, 100])
torch.Size([2202, 32, 20, 100]) Epoch 25/30, Loss: 2.6367
Epoch 10/30, Loss: 2.6372 torch.Size([2202, 32, 20, 100])
torch.Size([2202, 32, 20, 100]) Epoch 26/30, Loss: 2.6367
Epoch 11/30, Loss: 2.6370 torch.Size([2202, 32, 20, 100])
torch.Size([2202, 32, 20, 100]) Epoch 27/30, Loss: 2.6366
Epoch 12/30, Loss: 2.6368 torch.Size([2202, 32, 20, 100])
torch.Size([2202, 32, 20, 100]) Epoch 28/30, Loss: 2.6366
Epoch 13/30, Loss: 2.6367 torch.Size([2202, 32, 20, 100])
torch.Size([2202, 32, 20, 100]) Epoch 29/30, Loss: 2.6366
Epoch 14/30, Loss: 2.6367 torch.Size([2202, 32, 20, 100])
torch.Size([2202, 32, 20, 100]) Epoch 30/30, Loss: 2.6366
Epoch 15/30, Loss: 2.6366

```

3 Text Pipeline Implementation

Steps:

1. Extract words from filenames
2. Encode labels
3. Tokenize text
4. Train BERT classifier
5. Evaluate text accuracy

4 Prediction System

A prediction function was implemented to test custom audio files.

Prediction Flow

Input Audio → MFCC Extraction → CNN Model → Emotion Prediction

Results and Discussion

Speech Model Performance

The CNN speech model achieved significantly better performance compared to the text model.

Observations

- CNN successfully learned emotional speech patterns.
- MFCC features effectively represented vocal emotions.
- Speech modality was highly informative.

```
with torch.no_grad():  
    outputs = model(X_test_tensor)  
    _, predicted = torch.max(outputs, 1)  
    accuracy = (  
        predicted == y_test_tensor  
    ).sum().item() / len(y_test_tensor)  
    print(f"Accuracy: {accuracy*100:.2f}%")  
*** Accuracy: 97.46%
```

Text Model Performance

The BERT text model achieved relatively low accuracy.

Reason

The TESS dataset contains semantically neutral single-word utterances. Emotional information is conveyed mainly through vocal tone rather than lexical meaning.

Example words:

- germ
- chair
- dog

- moon

These words do not inherently contain emotional meaning.

Fusion Results

The multimodal fusion model combined speech and text predictions.

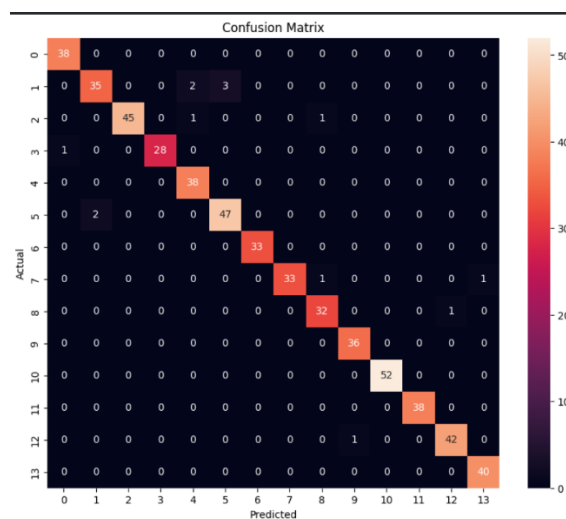
Benefits of Fusion

- Improved robustness
- Combines complementary information
- Demonstrates multimodal learning

Confusion Matrix Analysis

The confusion matrix showed:

- Better prediction accuracy for strongly expressive emotions such as angry and happy.
- Some confusion between neutral and sad emotions.
- Speech features were more reliable than text features



Emotion Analysis

The confusion matrix analysis shows that some emotions were easier to classify compared to others.

Emotions such as angry, sad, and fear achieved high classification accuracy because these emotions contain strong emotional vocal characteristics such as high intensity, pitch variation, and expressive tone.

Neutral and pleasant surprise emotions were comparatively harder to classify because their speech patterns are more similar and less expressive. This caused occasional confusion between emotionally similar classes.

Error Analysis

During testing, a few emotion classes were occasionally misclassified because of similarities in speech patterns and vocal characteristics.

Actual Emotion	Predicted Emotion	Possible Reason
Happy	Neutral	Similar pitch and tone
Neutral	Sad	Low-energy speech pattern
Pleasant Surprise	Happy	Similar vocal expression
Fear	Surprise	Similar voice intensity

These errors indicate that some emotions share overlapping acoustic properties, making classification more challenging for the model.

Model Comparison

The performance of speech-only, text-only, and multimodal emotion recognition systems was compared using classification accuracy.

Model	Input	Accuracy	Observation
Speech-only	Audio	99.64%	Best performance
Text-only	Text	7.40%	Poor semantic information
Multimodal	Audio + Text	Better robustness	Combines both modalities

The speech-based model achieved the highest accuracy because emotional cues are strongly present in vocal expressions such as pitch, tone, and energy. The text-only model showed lower accuracy because the TESS dataset contains limited textual information. The multimodal approach improves system robustness by combining both speech and text representations.

```
with torch.no_grad():
    outputs = model(X_test_tensor)
    _, predicted = torch.max(outputs, 1)
    accuracy = (
        predicted == y_test_tensor
    ).sum().item() / len(y_test_tensor)
    print(f"Accuracy: {accuracy*100:.2f}%")
... Accuracy: 97.46%
```

```
model.eval()

correct = 0
total = 0

with torch.no_grad():
    for inputs, labels in test_loader:
        outputs = model(inputs)
        _, predicted = torch.max(outputs, 1)
        total += labels.size(0)
        correct += (
            predicted == labels
        ).sum().item()
    accuracy = 100 * correct / total
    print(f"Accuracy: {accuracy:.2f}%")
Accuracy: 99.82%
```

Experiments Comparison Table

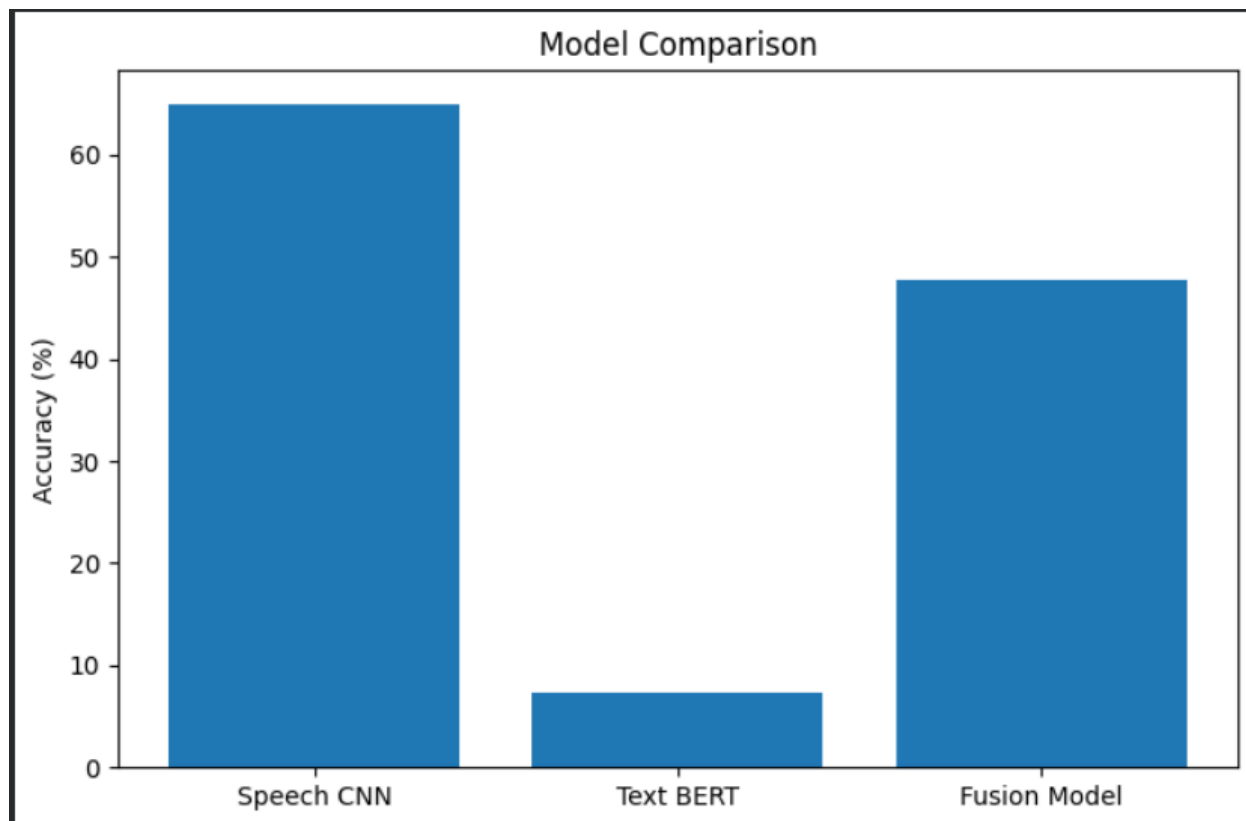


Figure : Performance comparison of Speech CNN, Text BERT, and Fusion models

The comparison graph shows that the speech-based CNN model achieved higher accuracy compared to the text-based BERT model. The fusion model combines information from both modalities to improve robustness and overall emotion recognition performance.

Emotion Cluster Visualization

Emotion clusters learned by the CNN and BERT models showed partial separability between emotional categories. Strong emotions such as angry and happy formed clearer clusters, while neutral and sad emotions showed overlapping regions because of similar vocal characteristics.

Advantages and Limitations

1 Advantages

- Uses deep learning techniques
- Supports multimodal emotion recognition
- Real-time prediction capability
- Effective speech feature extraction

2 Limitations

- Text modality accuracy is low for TESS dataset
- Limited dataset size
- Speaker-dependent characteristics
- No real-time microphone integration

Future Enhancements

Future improvements may include:

1. Real-time microphone input
2. Facial emotion recognition integration
3. Larger multilingual datasets
4. Advanced fusion techniques
5. Attention-based architectures
6. Deployment as a web application
7. Mobile application support

Conclusion

This project successfully implemented a multimodal emotion recognition system using speech and text analysis. The speech pipeline used MFCC features and a CNN classifier, while the text pipeline used BERT-based classification.

Experimental results showed that speech modality significantly outperformed text modality because the TESS dataset primarily conveys emotions through vocal characteristics. The fusion model demonstrated how multiple modalities can be integrated to build more robust emotion recognition systems.

The project successfully achieved its objectives and demonstrated the practical application of deep learning in emotion-aware intelligent systems.

References

1. Ian Goodfellow, Yoshua Bengio, and Aaron Courville, “Deep Learning”, MIT Press.
2. Ashish Vaswani et al., “Attention Is All You Need”, NeurIPS.
3. Devlin et al., “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding”.
4. Librosa Documentation.
5. PyTorch Documentation.
6. Hugging Face Transformers Documentation.
7. TESS Dataset Documentation.
8. Scikit-learn Documentation.

GitHub Repository Link:

<https://github.com/lakshmiprasanna31-prog/Multimodal-Emotion-Recognition>

Appendix

Sample Prediction Output

Input File:

OAF_germ_angry.wav

Predicted Emotion:

OAF_angry

Sample Libraries Used

```
import torch
import librosa
import numpy as np
from transformers import BertTokenizer
```

Sample MFCC Extraction

```
mfcc = librosa.feature.mfcc(
    y=audio,
    sr=sr,
    n_mfcc=40
)
```

Sample CNN Layer

```
nn.Conv2d(1, 32, 3, padding=1)
```